

Hao Wu

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PROFILE

I am a final-year Ph.D. candidate in Software Engineering at the Institute of Software, Chinese Academy of Sciences & University of Chinese Academy of Sciences (UCAS), supervised by [Prof. Naijun Zhan](#).

My research interests lie in *formal verification/synthesis* and *advanced topics in theoretical computer science*. Specifically, I investigate methodologies for combining symbolic computation with numerical optimization to advance the scalability of formal reasoning for infinite-state systems.

EDUCATION

- **Institute of Software, University of Chinese Academy of Sciences (UCAS)** Sep. 2020 - June. 2026 (planned)
Ph.D. Candidate in Software Engineering at Key Laboratory of System Software Beijing, China
◦ Advisor: [Prof. Naijun Zhan](#)
- **(Short-Term) The Hong Kong University of Science and Technology (HKUST)** Mar. 2024 - May. 2024
PG Visiting Internship Students (Research) Hong Kong, China
◦ Advisor: [Prof. Amir Goharshady](#)
- **University of Chinese Academy of Sciences (UCAS)** Sep. 2016 - June. 2020
B.Sc. (with honor) in Computer Science. Beijing, China
◦ Advisor: [Prof. Xiaoming Sun](#)

ACADEMIC SERVICES

- **External Reviewer.**
 - Conferences: HSCC/ICCPs'26, TACAS'25, HSCC'25, CAV'24, FM'24, iFM'24, ISSAC'23, CDC'23, TACAS'23, ChinaSoft'23, ICTAC'22
 - Journals: TAC, Automatica, Journal of Symbolic Computation (JSC), TCAD.
- **Teaching Assistant.**
 - 2023 Spring. Foundations of Theoretical Computer Science, B.Sc., UCAS

AWARDS

- First Class Ph.D. Scholarship in UCAS (2024, 2025).
- Outstanding Student in UCAS (2022).
- Distinguished B.Sc. Thesis Award in UCAS (2020).

PUBLICATIONS

By default, the author order (primarily the *first* author) is determined *by contribution*. Names marked with an asterisk (*) denote *equal contribution*, and names marked with a dagger (†) indicate *alphabetical order*.

Research Papers.

1. Yihao Yin, **Hao Wu**, Wan Liu, Shuling Wang, Xiong Xu, Wang Lin, Fanjiang Xu, and Naijun Zhan. **Formal Design of Safety-Critical Systems with MARS**. Journal of Systems Architecture, JSA, 2026 (to appear).
2. **Hao Wu***, Jiyu Zhu*, Amir Goharshady, Jie An, Bican Xia, and Naijun Zhan. [Quantifier Elimination Meets Treewidth](#). International Conference on Tools and Algorithms for the Construction and Analysis of Systems, TACAS, 2026 (to appear).
3. **Hao Wu**, Qiuye Wang, Bai Xue, Naijun Zhan, Lihong Zhi, and Zhi-Hong Yang. [Synthesizing Invariants for Polynomial Programs by Semidefinite Programming](#). ACM Transactions on Programming Languages and Systems, TOPLAS, 2025.

4. Xiong Xu, Jean-Pierre Talpin, Shuling Wang, **Hao Wu**, Bohua Zhan, Xinxin Liu, and Naijun Zhan. **HpC: A Calculus for Hybrid and Mobile Systems**. In the ACM Conference on Object-Oriented Programming, Systems, Languages, and Applications, OOPSLA, 2025.
5. Guanhua Lin, **Hao Wu**[†], and Naijun Zhan. **Barrier Certificate Synthesis via Interpolation and Difference-of-Convex Programming**. In the Proc. of Colloquium on Design and Verification of Cyber-Physical Systems: From Theory to Applications, Festschrift of Prof. Martin Fränzle's 60th birthday, 2025.
6. **Hao Wu**, Yu-Fang Chen, Zhilin Wu, Bican Xia, and Naijun Zhan. **A decision procedure for string constraints with string/integer conversion and flat regular constraints**. Acta Informatica, 2024.
7. **Hao Wu**^{*}, Jie Wang^{*}, Bican Xia, Xiakun Li, Naijun Zhan, and Ting Gan. **Nonlinear Craig Interpolant Generation Over Unbounded Domains by Separating Semialgebraic Sets**. In the International Symposium on Formal Methods, FM, 2024.
8. **Hao Wu**, Shenghua Feng, Ting Gan, Jie Wang, Bican Xia, and Naijun Zhan. **On Completeness of SDP-Based Barrier Certificate Synthesis over Unbounded Domains**. In the International Symposium on Formal Methods, FM, 2024.
9. Yihao Yin, **Hao Wu**, Shuling Wang, Xiong Xu, Fanjiang Xu, and Naijun Zhan. **The Design of Intelligent Temperature Control System of Smart House with MARS**. In the CCF International Symposium on Dependable Software Engineering Theories, Tools and Applications, SETTA, 2024.
10. Xiaohui Bei, Xiaoming Sun, **Hao Wu**[†], Jialin Zhang, Zhijie Zhang, and Wei Zi. **Cake Cutting on Graphs: A Discrete and Bounded Proportional Protocol**. In the ACM-SIAM Symposium on Discrete Algorithms, SODA, 2020.

Survey Papers.

1. Hao Wu, Shenghua Feng, Lihong Zhi, and Naijun Zhan. **Optimization-based Invariant Generation for Cyber-Physical Systems** (in Chinese). China Basic Science, 2026. (To Appear)
2. Shuling Wang, Bohua Zhan, Huanhuan Sheng, **Hao Wu**, Shicheng Yi, Lingtai Wang, Xiangyu Jin, Bai Xue, Jinghui Li, Shuang-Qing Xiang, Zhan Xiang, and Bifei Mao. **Survey on Requirements Classification and Formalization of Trustworthy Systems** (in Chinese). Journal of Software, 2022.

PROJECTS AND GRANTS

- Participant in NSFC Key International Joint Research Program (China, France): Research on Design Techniques for Safety-Critical Cyber-Physical Systems in the Context of Mobile Computing
- Participant in National Key R&D Program of China under grant No. 2022YFA1005101
- Participant in Natural Science Foundation of China under grant No. 62192732.

REFERENCES

1. **Prof. Naijun Zhan**
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Peking University (PKU)
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Relationship: Advisor
2. **Prof. Amir Goharshady**
Associate Professor, Department of Computer Science
University of Oxford
Email: amir.goharshady@cs.ox.ac.uk
Relationship: Advisor of HKUST Internship